Science Week 4: moments (written)

Ember · Moments and levers, with a real AQA GCSE Physics question · about 15 minutes · pencil and ruler

Date: _____

Skill: the Because Chain. One mark per link. For calculations, write the equation, then the numbers, then the answer with its unit.

[3 marks]

1. A uniform plank is balanced on a pivot at its centre. A 60 N weight is placed 0.50 m to the left of the pivot. Calculate where an 80 N weight must be placed on the right of the pivot to balance the plank. Give your answer in metres.

Working

Answer Distance = _____ m

Plan the chain for question 2 (one short phrase per box)

Where is the force?	So the distance...	The moment needed...	So the force...

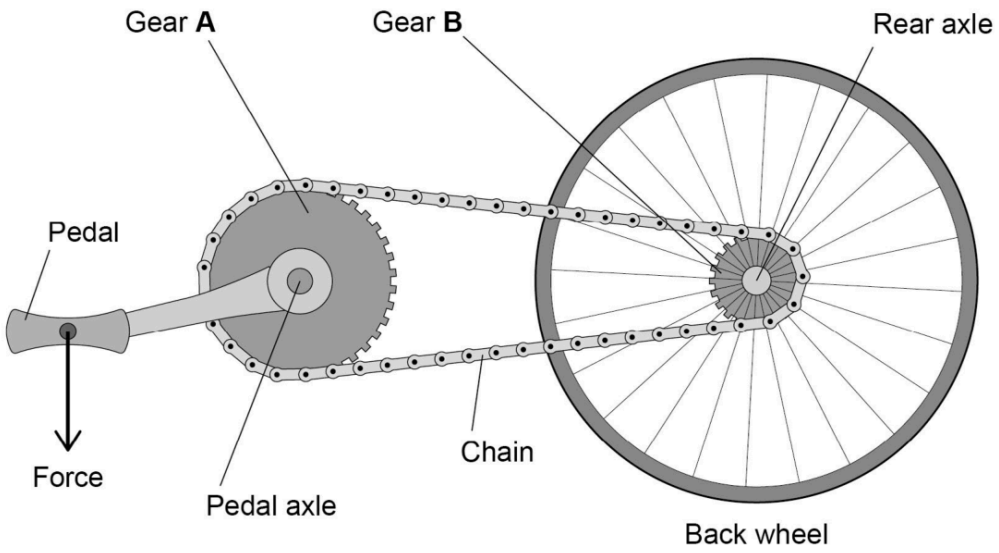
[2 marks]

2. Scissors with long handles cut thick card more easily than scissors with short handles. The pivot is the screw in the middle. Explain why.

AQA Physics 8463/2H, November 2021, Q07.3 **[2 marks]**

Figure 13 shows the gears on the bicycle.

Figure 13



November 2021, Paper 2, Figure 13

Describe how the force on the pedal causes a moment about the rear axle.

Self-check (5 min)

□ Question 1: I wrote the anticlockwise moment, then made the clockwise moment equal to it.

□ Question 2: I said the distance from the pivot is greater, **then** that a smaller force is needed.

□ The bicycle question: I described two steps: the pedal turns gear A, then the chain turns gear B.

□ I used the word **moment** and the word **pivot** (or axle).

□ Reflection: which link did I nearly leave out?

Question 1 (3 marks)

- anticlockwise moment = 60 × 0.50 = 30 (N m) **[1]**
- 80 × d = 30 **[1]**
- d = 0.375 m (allow 0.38 m) **[1]**

The heavier weight sits closer to the pivot. Score: ____ / 3

Question 2 (2 marks)

- the (perpendicular) distance from the pivot to the force (the hands) is greater with long handles **[1]**
- so a smaller force is needed (to produce the same moment) **[1]** *(only if the first mark is scored)*

Marked the same way as AQA June 2019 Q10.1 (the lorry ramp). Score: ____ / 2

AQA mark scheme: November 2021, Q07.3 (2 marks)

- (the force on the pedal) causes a moment about the pedal axle **[1]**
- which causes a force on the chain (which causes a moment about the rear axle) **[1]** *(allow gear B for chain)*

It is a chain again: pedal → moment about the pedal axle → force on the chain → moment about the rear axle. Score: ____ / 2

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